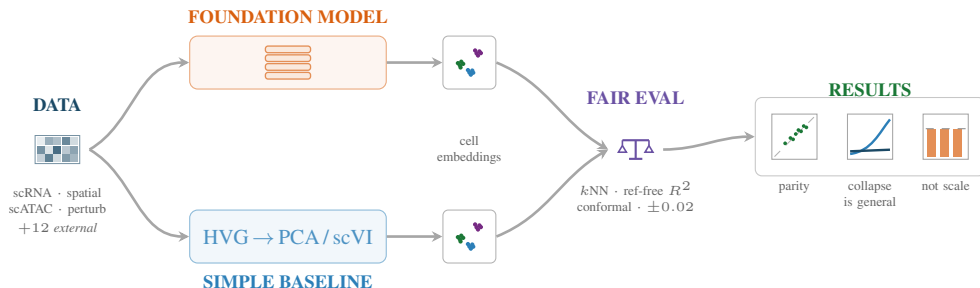


A**B**

Two camps: *simple beats FM* (skeptics) vs. *FM wins* (boosters) — leaderboards can't adjudicate

fair re-check ($k\text{NN} + \text{ref-free } R^2_{\text{expr}}$)

dose-response #1
vocabulary

dose-response #2
batch-shift

dose-response #3
spatial

1 Parity — no advantage, no deficit (= PCA on matched atlases; $+0.0025, \pm 0.02$)
apparent win = label circularity + vocab. threshold

2 General collapse — an exchangeability failure, not FM-specific ($\rho=0.82$; scATAC flat)

3 Not scale — set by task and metric, not model size (104→316M: no effect)

Deliverable — a reusable, falsifiable evaluation instrument, not a verdict on FMs